Journal of Luminescence 263 (2023) 120130

Contents lists available at ScienceDirect

Journal of Luminescence

journal homepage: www.elsevier.com/locate/jlumin

Full Length Article

Visible upconverted and NIR luminescence properties of Er^{3+} and Er^{3+}/Yb^{3+}-doped $Lu_3Sc_2Ga_3O_{12}$ nanocrystals

Ramadevi N [a], Praveena R [b,*], Venkatramu V [c], Basavapoornima Ch [d], Lavín V [e], Joshi B.D [f], Raja Rao N S S V [b]

[a] Central Department of Physics, Tribhuvan University, Kirtipur, Kathmandu, Nepal
[b] Department of Physics, Gayatri Vidya Parishad College of Engineering (A), Visakhapatnam-530 048, India
[c] Department of Physics, Krishna University, Dr. MR Appa Row college of Post Graduate Studies, Nuzvid- 521 201, India
[d] Institute of Aeronautical Engineering, Hyderabad-500 043, India
[e] Departamento de Física, MALTA-Consolider Team, and IUdEA, Universidad de La Laguna, Apartado de Correos 456, E-38200 San Cristóbal de La Laguna, Santa Cruz de Tenerife, Spain
[f] Department of Physics, Siddhanath Science Campus, Tribhuvan University, Mahendranagar, Nepal

ARTICLE INFO

Keywords:
Er^{3+}
Yb^{3+}
LuSGG
Nanocrystals
Photoluminescence
Upconversion

ABSTRACT

Trivalent erbium and ytterbium embedded in $Lu_3Sc_2Ga_3O_{12}$ nanocrystals in the garnet phase were synthesised employing the citrate sol-gel method. The crystalline phase and size, grain size, and effect of ion concentration and laser pump power on the photoluminescence properties of lanthanide ions were systematically characterised. XRD results showed that the present phosphor was in the cubic garnet phase with an average crystallite size of 27 nm and a lattice constant of 12.325 Å. Morphology images revealed the particles to be nearly spherical and agglomerated. Partial energy level diagrams have been constructed using diffused reflectance and excitation spectra. Stokes and anti-Stokes photoluminescence spectra were observed under 980 nm laser excitation. The near-infrared (NIR) photons (980 nm) were successfully converted into visible radiation. Upconverted luminescence spectra consisted of two strong peaks located in the green and red regions, corresponding to the $^2H_{11/2}$,$^4S_{3/2} \rightarrow {}^4I_{15/2}$ and $^4F_{9/2} \rightarrow {}^4I_{15/2}$ transitions of the Er^{3+} ion. The intensity ratio of red to green emission bands increased with increasing Er^{3+} or Yb^{3+} concentrations as well as with the pump power of laser. Power-dependent photoluminescence spectra revealed that there were two photons involved in the upconversion mechanism. The decay profiles of the thermally coupled $^2H_{11/2}$,$^4S_{3/2}$ levels of Er^{3+} showed a non-exponential nature accompanied by a shortening of lifetime as the concentration increased. CIE colour co-ordinates were located in the green region and shifted to the yellow region with increasing Yb^{3+} ion concentration. NIR emission spectra exhibited an intense peak at around 1535 nm, attributed to the $^4I_{13/2} \rightarrow {}^4I_{15/2}$ transition of Er^{3+} ion, the intensity of which was observed to increase with the incorporation of Yb^{3+} ions up to 5 mol%, confirming the existence of Yb^{3+}-to-Er^{3+} energy transfer processes. Therefore, the present phosphors can be used for photovoltaic cells, solid-state light emitting diodes, and display devices.

1. Introduction

Lanthanide (Ln)-doped materials have been widely used in various optical devices like photovoltaic cells, lamps, light emitting diodes, light converters, optical thermometers, bio-imaging, and display technologies owing to their unique characteristics such as narrow emissions and excitation bands originating from *f-f* electronic transitions [1,2]. Among trivalent lanthanide (Ln^{3+}) ions, Er^{3+} ions are commonly used as activators because of their luminescence in green and red spectral regions arising from the ($^2H_{11/2}$,$^4S_{3/2}$)$\rightarrow {}^4I_{15/2}$ and $^4F_{9/2} \rightarrow {}^4I_{15/2}$ transitions. In addition, Er^{3+} ions can exhibit infrared-to-visible photon energy conversion processes, usually known as upconversion, due to their ladder-type metastable energy levels, although their weak absorption at 980 nm provides low luminescence efficiency that limits their practical applications [3–5]. Fortunately, the Yb^{3+} ion is often considered as sensitizer to the Er^{3+} ion owing to the spectral overlap between the $^2F_{5/2} \rightarrow {}^2F_{7/2}$ transition of Yb^{3+} and the $^4I_{15/2} \rightarrow {}^4I_{11/2}$ transition of Er^{3+} at 980 nm. Moreover, the Yb^{3+} ion has only one excited state at 10,000

* Corresponding author.
E-mail address: praveena@gvpce.ac.in (P. R).

https://doi.org/10.1016/j.jlumin.2023.120130
Received 1 May 2023; Received in revised form 31 July 2023; Accepted 11 August 2023
Available online 21 August 2023
0022-2313/

cm^{-1} [6] that is suitable for pumping by the commercial laser diodes and helps to avoid excited state absorption. In addition, Yb^{3+} has a much larger absorption cross-section for 980 nm near-infrared (NIR) radiation than Er^{3+} ions. Therefore, efficient energy transfer from Yb^{3+} to Er^{3+} ions could be expected, which can improve the efficiency of the upconverted luminescence [4,7]. Normally, a high concentration of Yb^{3+} ions is needed to maintain fast energy migration between donors, which enables efficient donor-to-acceptor energy transfer processes. Moreover, Yb^{3+} ions also play a key role in adjusting the intensity proportion of green and red emissions as a sensitizer. Hence, Er^{3+}/Yb^{3+}-doped upconversion phosphors are most desirable as they play a key role in bio-imaging, crime scene analysis, anticounterfeiting, optical coherence tomography, photodynamic therapy, temperature sensing, etc., due to their excellent optical properties, less damage to biological tissue, strong tissue penetration, low autofluorescence, low toxicity, and large anti-Stokes shifts [8–11]. However, the upconversion efficiency is also influenced by the sample chosen. A host material with low phonon energies is favourable for obtaining efficient upconversion luminescence. Therefore, a suitable host material must be selected.

Recently, garnet nanocrystals have gained great attention due to their excellent physical and chemical properties, such as high thermal conductivity, high density, high transparency from the UV to the IR, hardness, good chemical stability, and relatively low phonon energy [12–14]. With well-known synthesis techniques, Ln^{3+}-doped nanocrystalline garnets allow tunable properties to be achieved as a function of crystalline size and shape [15,16]. Garnets have already shown that they are excellent host materials for Ln^{3+} ions. For instance, Er^{3+}-doped $Y_3Ga_5O_{12}$ nano-garnet showed relatively high sensitivity than other samples and was proposed as a temperature sensor in biological systems [17]. Photoluminescence properties of Er^{3+}/Yb^{3+} co-doped $Y_3Ga_5O_{12}$ nano-garnets were investigated for their potential applications as nanothermometers and nanoheaters in the first biological window [18]. Monteseguro et al. [19], reported Stokes and upconverted luminescence in Er^{3+}/Yb^{3+} co-doped $Y_3Ga_5O_{12}$ (YGG) nano-garnets. However, it has been proven that Lu^{3+}-based compounds show higher luminescence intensities than the corresponding Y^{3+} and Gd^{3+}-based compounds [20, 21]. Among all the garnets, $Lu_3Ga_5O_{12}$ is an excellent garnet host for the Ln^{3+} ions that can show upconverted luminescence. Rathaiah et al. [3], observed infrared-to-visible light conversion in Er^{3+}/Yb^{3+} co-doped $Lu_3Ga_5O_{12}$ (LuGG) nano-garnets. With Sc^{3+} ions replacing part of the Ga^{3+} ion in Ln^{3+}:LuGG, a new class of Ln-doped mixed garnet $Lu_3Sc_x Ga_{5-x}O_{12}$ (LuSGG) can be formed. From the literature, it is clear that LuSGG crystal has a broad emission spectrum and a small emission cross-section [22,23]. Therefore, in the present work, NIR and infrared-to-visible upconverted luminescence of Er^{3+} and Er^{3+}/Yb^{3+}-doped $Lu_3Sc_2Ga_3O_{12}$ nano-garnets under 980 nm excitation have been studied as a function of Er^{3+} or Yb^{3+} ion concentrations as well as the laser pump power. To the best of the author's knowledge, there is no report on the present study.

2. Experimental techniques

Er^{3+} and Er^{3+}/Yb^{3+} co-doped lutetium scandium gallium nano-garnets of composition $Lu_{3-x}Sc_2Ga_3O_{12}$:xEr^{3+} (where x = 0.03, 0.15, and 0.30) and $Lu_{3-0.03-y}Sc_2Ga_3O_{12}$:$0.03Er^{3+}/yYb^{3+}$, where y = 0.03, 0.15, and 0.30 (representing x and y as 1, 5 and 10 mol%) were prepared by the sol-gel method [24]. For convenience, hereafter these samples are labelled as LuSGG:1Er, LuSGG:5Er, LuSGG:10Er, LuSGG:1Er1Yb, LuSGG:1Er5Yb, and LuSGG:1Er10Yb. All chemicals have been procured from Sigma-Aldrich with high purity (≥99.9%). The appropriate amounts of raw materials of lutetium (III) nitrate hexahydrate, erbium (III) nitrate pentahydrate, ytterbium (III) nitrate pentahydrate, scandium (III) nitrate hydrate, and gallium (III) nitrate hydrate were dissolved in 25 ml of 1 M HNO_3 while stirring at 70 °C for 3 h. Later, citric acid and PEG were added during stirring, maintaining the metal:citric acid:PEG ratio of 1:2:8. Samples were dried at 90 °C for 36 h to obtain the gel phase. The obtained gel was first heated at 500 °C for 4 h in order to remove the extra nitrates and organic compounds, and then fired at 950 °C for 16 h to obtain nanocrystalline particles.

The structure of the samples was studied by X-ray diffraction (XRD) patterns recorded in the range of 2θ = 15-75°, with a step size of 0.02° by using the X-ray diffractometer (RIGAKU, Miniflex-600) with Cu k_α radiation (1.5406 Å). Scanning electron microscope (CARL ZEISS EVO MA15 SEM) images were used to determine the morphology of the samples. Visible-NIR diffuse reflectance spectra in the range from 300 to 1700 nm were collected with a spectrophotometer (Varian Cary 6000i) equipped with an integrating sphere. Excitation, emission, and decay measurements were done using a UV-VIS-NIR spectrometer (Edinburgh Instruments FLS 980) with a 980 nm laser diode as an excitation source. All the measurements were carried out at room temperature.

3. Results and discussion

3.1. Structure and morphology

XRD patterns of LuSGG1Er and LuSGG1Er1Yb powder samples, taken in the 2θ range from 15 to 75° and shown in Fig. 1(a), contain sharp peaks that represent a well crystallised, single cubic phase of space group *Ia-3d* (No. 230), which matches well with the standard JCPDS card values (#54–1252). This shows that the Ln^{3+} ions are properly incorporated into the LuSGG crystal without distorting the structure. From Fig. 1(a), it is observed that co-doping with the Yb^{3+} ion has no influence on the shapes and positions of diffraction peaks. In both samples, two non-indexed peaks were located at 21.4 and 30.5° in the XRD patterns that are due to a small excess of Sc_2O_3, matched with JCPDS no. #74–1210 [25], with no further consequences on the optical properties of Ln^{3+} ions. Similar results were observed in Nd^{3+}:$Y_3Sc_2AlGa_2O_{12}$ nano-powders [26] and Yb^{3+}:LuSGG nanocrystals [27]. A careful and systematic phase analysis of the obtained X-ray diffraction patterns was carried out using Materials Analysis Using Diffraction (MAUD) software. The obtained results clearly show that the $Lu_3Sc_2Ga_3O_{12}$ phase is above 90% and the Sc_2O_3 contribution is small, i.e., less than 10%. The refinement parameters are given in Table 1, and the fittings are shown in Fig. 1(b) and (c) for LuSGG:1Er and LuSGG:1Er1Yb, respectively.

In general, LuSGG nano-garnet crystals are composed of a three-dimensional network of GaO_4 tetrahedra and GaO_6 octahedra linked by sharing oxygen ions at the corners of the polyhedron [21], in which part of these Ga^{3+} ions are replaced by Sc^{3+} ones. Chains of these polyhedra are extended along the three crystallographic directions and create dodecahedral cavities that can be occupied by the Ln^{3+} ions [28]. In the present case, Er^{3+} and Yb^{3+} ions are expected to predominately enter the distorted dodecahedra sites, replacing Lu^{3+} ions due to ionic size considerations, and without charge compensation.

The average crystallite size (D) is estimated using Debye-Scherer's formula,

$$D\frac{k\lambda}{\beta\cos\theta} \tag{1}$$

where λ is the wavelength of X-ray; β is the full width at half maximum (FWHM); θ is the angle of diffraction peak; and k is the shape factor (0.89). The average crystallite size, inter-planar spacing, and lattice constant for the present samples were found to be 27 nm, 2.756 Å, and 12.325 Å, respectively. The lattice constant of LuSGG:1Er1Yb is found to be larger compared to LuGG:1Er1Yb (12.193 Å) [3], because part of the Ga^{3+} ions, with an ionic radius of 62 p.m., are replaced by Sc^{3+} ions with a higher ionic radius (74.5 p.m.). On the other hand, the lattice constant in LuSGG:1Er1Yb is lower compared to the YSGG crystal (12.42 Å) [29], indicating that the Ln^{3+} ions experience a tighter lattice in the LuSGG, resulting in stronger crystal-field interactions with oxygen ligands. This implies that a larger Stark splitting of the energy-level manifold would

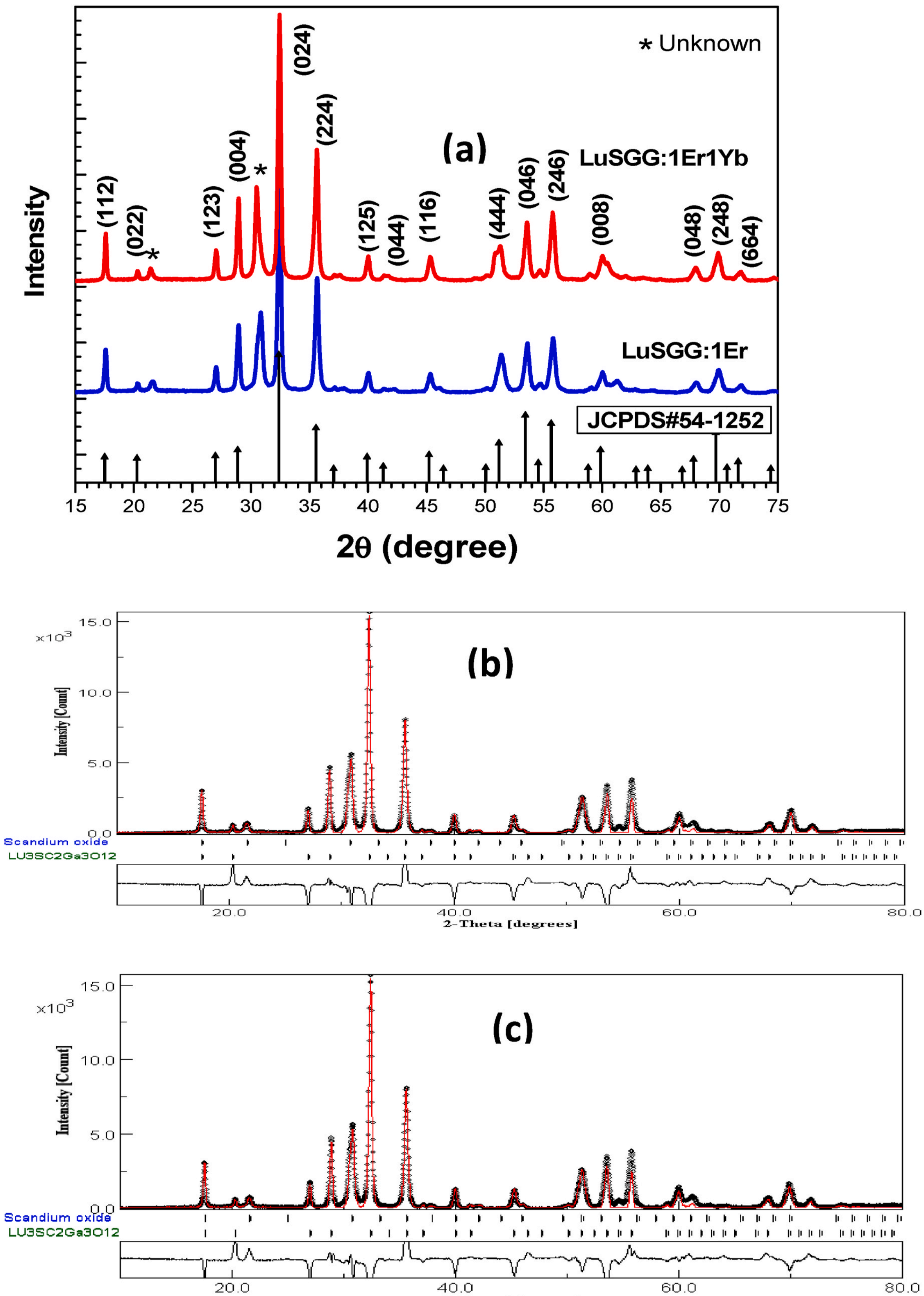

Fig. 1(a). XRD patterns of the LuSGG:1Er and LuSGG:1Er1Yb nanocrystals and (b&c) MAUD refinement for LuSGG:1Er and LuSGG:1Er1Yb nanocrystals.

Table 1
Refinement parameters for LuSGG:1Er and LuSGG:1Er1Yb nanocrystals.

Sample	χ^2	R_{wp} (%)	R_b(%)	R_{exp} (%)	Phase %		Lattice parameter (a = b = c) (Å)
					$Lu_3Sc_2Ga_3O_{12}$	Sc_2O_3	
LuSGG:1Er	0.65	2.2811	1.6977	12.9065	93.43	6.57	12.3332
LuSGG:1Er1Yb	1.04	3.6241	2.4378	9.9146	91.29	8.71	12.3444

tend to produce lower thermal occupation factors of both the upper and lower laser levels, which benefits population inversion and a reduced lasing threshold at room temperature [30]. Substitution of Y^{3+} ions for Lu^{3+} in host materials is more appropriate for doping Er^{3+} ions, since the very small variance of atomic mass between Lu^{3+} (175.0 g/mol) and Er^{3+} (167.3 g/mol)−for Y^{3+}, it is overweighed (88.9 g/mol) −results in a lower phonon energy [31].

SEM images of LuSGG:1Er and LuSGG:1Er1Yb nanocrystals that were used to investigate the morphological features of the prepared nanocrystals are shown in Fig. 2. The particles appear to be nearly spherical and crystallised in an inhomogeneous morphology, showing a tendency to agglomeration and grain growth. Grain size varies between 70 and 160 nm, due to the agglomeration during synthesis, produced by established cohesive forces between the grains [32,33]. Majorly, van der Waals force and electrostatic interaction cause the force between inter-particles, which is mainly dependent on the atomic and molecular properties of the surface atoms, morphology, and proximity. The complexity of cohesive forces between grains is significantly increased by these factors.

3.2. Diffuse reflectance and excitation

Diffuse reflectance spectra (DRS) of the LuSGG:1Er and LuSGG:1Er10Yb nanocrystals measured in the range between 300 and 1700 nm are displayed in Fig. 3(a) and (b). The sharply structured peaks in the DRS confirm the substitution of Ln^{3+} ions in the LuSGG crystalline garnet network. These peaks are attributed to intra-configurational *4f-4f* electronic transitions between the ground and excited state multiplets of the Er^{3+} ions. The spectra consist of 11 absorption peaks centred at 367, 380, 407, 442, 450, 488, 522, 542, 655, 789, 972, and 1534 nm, which are assigned to $^4I_{15/2} \rightarrow {}^2G_{9/2}$, $^2G_{11/2}$, $^2H_{9/2}$, $^4F_{3/2,5/2}$, $^4F_{7/2}$, $^2H_{11/2}$, $^4S_{3/2}$, $^4F_{9/2}$, $^4I_{9/2}$, $^4I_{11/2}$, and $^4I_{13/2}$ transitions, respectively. All these transitions are electric-dipole in nature except for the $^4I_{13/2}$ level of the Er^{3+} ion, which also contains a magnetic-dipole contribution. The spectra are similar to those of other Er^{3+}/Yb^{3+}-doped systems [19,34,35].

With the incorporation of Yb^{3+} ions into the LuSGG:1Er, an intense broad absorption band is noticed in the region between 890 and 1050 nm due to the overlapping of two absorption peaks that are attributed to the $^4I_{15/2} \rightarrow {}^4I_{11/2}$ transition of the Er^{3+} ion and the $^2F_{7/2} \rightarrow {}^2F_{5/2}$ transition of the Yb^{3+} ion. The main peak is centred at 973 nm with a FWHM of 9.5 nm, which is comparable with those found for LuSGG:20Yb (9.05 nm) [36] and LuSGG:1Tm10Yb (9.65 nm) [37]. As given in Ref. [36], the observed peaks are attributed to the transitions between Stark levels of the Yb^{3+} ion. The ground state $^2F_{7/2}$ splits into four Stark levels (1, 2, 3, and 4) and the excited state splits into three ones (5, 6 and 7). Hence, peaks at 896, 931, 973, 944, and 1024 nm correspond to the transitions between 1 → 7, 1 → 6, 1 → 5, 3 → 5, and 4 → 5 of the $^2F_{7/2}$ and the $^2F_{5/2}$ multiplets, respectively. Therefore, the present spectra confirm the higher absorption cross-section at 980 nm of the LuSGG:Er^{3+} sample containing Yb^{3+}. Furthermore, the band located in the 900–980 nm region matches well with the emission band of a low-cost commercial InGaAs diode laser, showing potential applications in display devices and solid-state lighting applications.

The excitation spectra of Er^{3+}/Yb^{3+} co-doped LuSGG nanocrystals were recorded in the wavelength range between 350 and 550 nm, monitoring the emission at 554 nm, and are shown in Fig. 4. The spectra consist of 7 peaks centred at around 357, 367, 380, 407, 451, 488, and 522 nm corresponding to various transitions from the ground state ($^4I_{15/2}$) of Er^{3+} to the excited levels $^2G_{7/2}$, $^2G_{9/2}$, $^2G_{11/2}$, $^2H_{9/2}$, $^4F_{3/2,5/2}$, $^4F_{7/2}$, and $^2H_{11/2}$ multiplets, respectively. As expected, the observed transitions in the excitation spectra were also found in the diffuse reflectance spectra shown in Fig. 3. Fig. 4 shows that the intensity of the peaks decreases with increasing Yb^{3+} ion concentration, probably due to the enhancement of energy back transfer (EBT) from Er^{3+} ions to Yb^{3+} ions via $^4S_{3/2}$, $^2F_{7/2} \rightarrow {}^4I_{13/2}$, $^2F_{5/2}$. However, the Stokes emission spectra at 980 nm can provide detailed information about this energy transfer, which is beyond the scope of this study.

3.3. NIR-to-visible upconverted luminescence

The optical properties of Ln^{3+} ions are sensitive to the environment, or crystalline site, in which they are located in the host matrix [38]. The crystal-field interaction acting on the Ln^{3+} ion causes the splitting of the free-ion $^{2S+1}L_J$ multiplets and the intra-configurational *4f-4f* electric-dipole transitions in the ultraviolet to NIR region. It is assumed that the Er^{3+} and Yb^{3+} ions substitute Lu^{3+} ion sites with D_2 symmetry in the LuSGG host. This symmetry lifts the degeneracy of the multiplets and gives rise to the (2J + 1)/2 Stark levels [39]. NIR-to-visible upconverted emission spectra of LuSGG:Er and LuSGG:Er/Yb nanocrystals were investigated under 980 nm laser excitation and are shown in Fig. 5(a) and (b), respectively. The sharp emission peak profiles in Fig. 5 confirm that the Ln^{3+} ions occupied the distorted dodecahedral sites with D_2

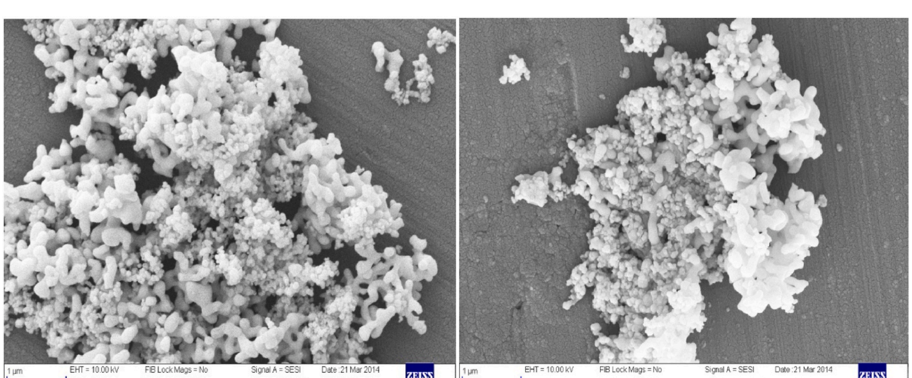

Fig. 2. SEM images of LuSGG:1Er and LuSGG:1Er1Yb nanocrystals.

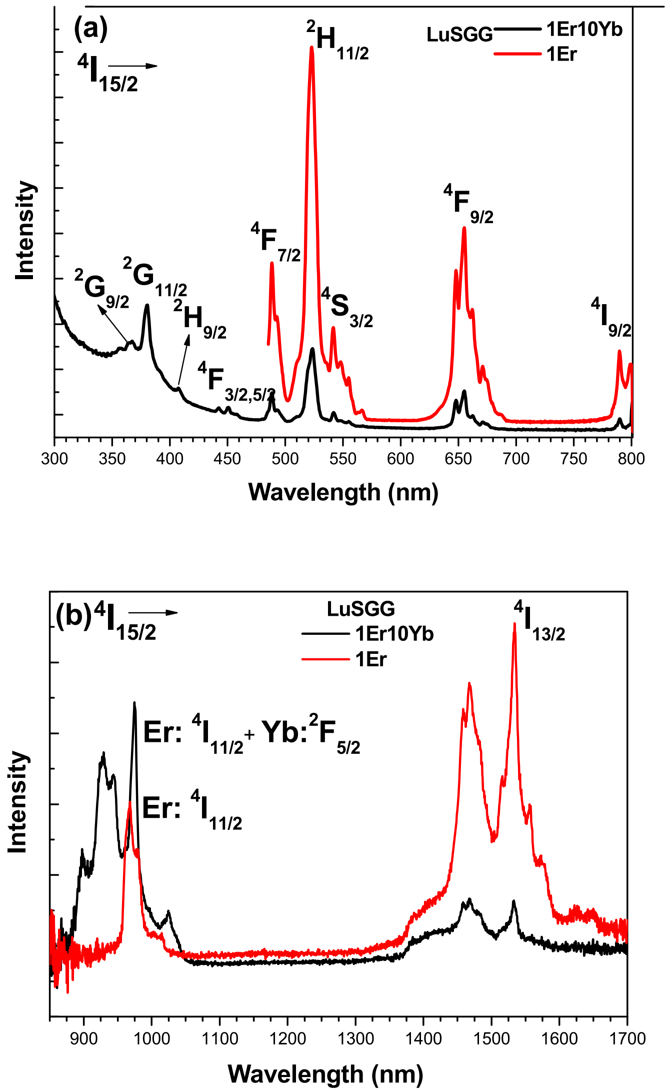

Fig. 3. Diffuse reflectance spectra of LuSGG:1Er and LuSGG:1Er10Yb nanocrystals in the (a) 300–800 nm and (b) 850–1700 nm region.

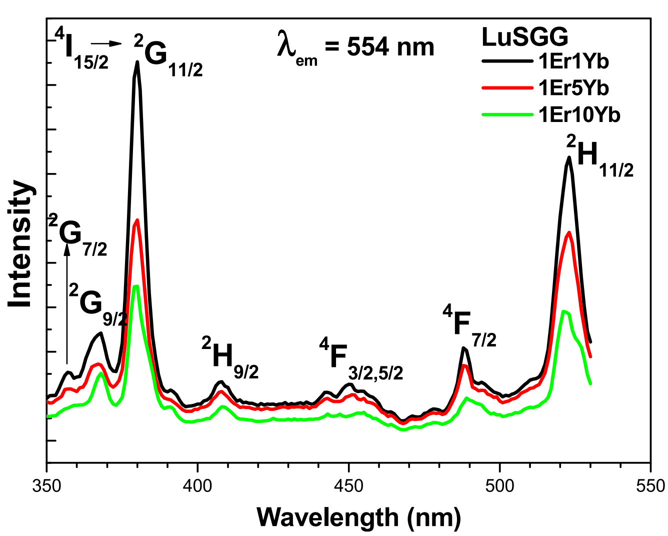

Fig. 4. Excitation spectra of LuSGG:Er/Yb nanocrystals with varying Yb^{3+} concentration.

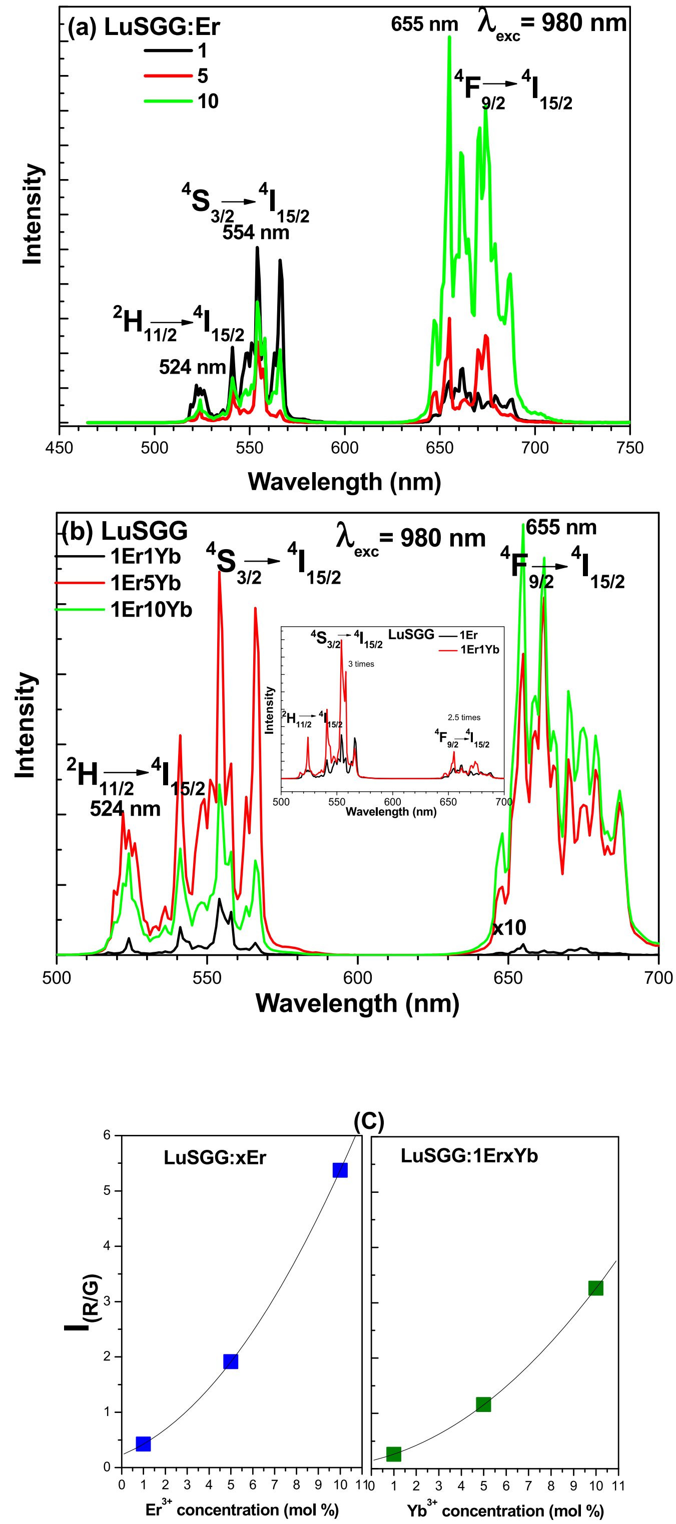

Fig. 5. Upconversion luminescence spectra of (a) LuSGG:Er and (b) LuSGG:Er/Yb at 676 mW power. (c) R/G ratio in LuSGG:Er and LuSGG:Er/Yb nanocrystals.

symmetry in the LuSGG nanocrystal. Upconverted luminescence spectra of both LuSGG:Er and LuSGG:Er/Yb consist of two distinct emission bands: one is green, extended in the 510–580 nm region and attributed to the ($^2H_{11/2}$,$^4S_{3/2}$) →$^4I_{15/2}$ transition, and the other is red, extended in the 630–720 nm range, and corresponds to the $^4F_{9/2}$ → $^4I_{15/2}$ transition of the Er^{3+} ion. The splitting observed in the green and red emissions is due to Stark splitting manifolds induced by the crystal-field effect around the central optically active ion. As can be seen from Fig. 5(a), the green luminescence intensity is slightly diminished at 5 mol% and then increased at higher concentrations, whereas the intensity of the red luminescence is enhanced gradually with increasing the concentration of Er^{3+} ions. Similarly, from Fig. 5(b), it is noticed that, with increasing Yb^{3+} ion concentration, the intensity of the green luminescence increases up to 5 mol% and then decreases, whereas the intensity of the red luminescence continues to increase even beyond 5 mol%. With the incorporation of Yb^{3+} ions, the intensity of the green emission is enhanced 3 times, whereas the intensity of the red emission is enhanced 2.5 times (see inset of Fig. 5(b)). This enhancement indicates that the Yb^{3+} ions are effectively pumped under 980 nm laser excitation due to the large contribution of energy transfer processes from Yb^{3+}-to-Er^{3+} ions. Fig. 5(c) displays the integrated red-to-green (R/G) intensity ratio in both the LuSGG:Er and LuSGG:Er/Yb nanocrystals. It is observed that the R/G ratio is raised from 0.42 to 5.37 with increasing Er^{3+} and from 0.26 to 3.26 with increasing Yb^{3+} ion concentration, and the enhancement is relatively high in LuSGG:Er nanocrystals.

In order to analyse the mechanisms behind the upconverted luminescence, the power-dependent luminescence spectra of LuSGG:Er/Yb with 980 nm excitation were measured. The change in laser power did not alter the overall shape and positions of the peaks except for their intensities, indicating that the present samples show good stability under laser excitation power. Both the green and red emission intensities are observed to increase with increasing pump power (not shown). However, the R/G ratio does not change with pump power, indicating that both the green and red luminescence intensities vary almost constantly with respect to the laser pump power. This means that both emissions are competing with each other.

It is also well-known that the upconverted luminescence intensity (I_{up}) is proportional to the nth power of the infrared excitation intensity (I_{IR}), i.e., $I_{up} \propto (I_{IR})^n$, where '$n$' is the number of infrared photons absorbed per one visible photon emitted [39]. Fig. 6 shows the Log(Intensity) vs. Log(power) graph, where the slope of this graph gives the 'n' value. From Fig. 6, it is noticed that the 'n' values for green and red emissions are 1.15 and 1.26, respectively, suggesting that there are two photons involved in the upconversion process. The deviation of the slope from the theoretical value of 2 is ascribed to saturation resulting from competition between the radiative relaxation probabilities of intermediate states and the upconversion probability [40–42]. Consequently, in such cases, the up-converted intensity should become constant or independent of pump intensity at very high pump power. Vijay Singh et al. [35], observed the same saturation effect in Er^{3+} and Er^{3+}/Yb^{3+}-doped Y_2O_3 phosphors for high pump powers. Mahalingam et al. [43] also observed the saturation phenomenon that depends upon excited pumping power in $Tm^{3+}/Yb^{3+}/Ho^{3+}$: $GdVO_4$ nanocrystals. Singh and Geusic [42] observed the saturation behavior of up-conversion in $NdCl_3$ single crystal for higher pump power.

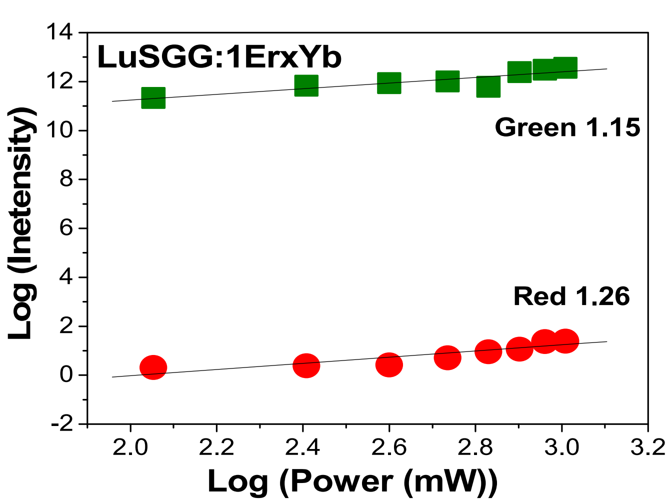

Fig. 6. Logarithmic plot between luminescence intensity and laser power.

It is often convenient to describe the upconversion processes in single-doped and co-doped LuSGG nanocrystals using partial energy level diagrams, as displayed in Fig. 7. In the case of a single-doped LuSGG:Er sample, the ground-state Er^{3+} ions are excited from $^4I_{15/2}$ to the higher $^4I_{11/2}$ level by absorbing a 980 nm photon through a ground-state absorption process (GSA). Then, a portion of the Er^{3+} ions relax to the $^4I_{13/2}$ level due to a non-radiative transition involving multiphonon interaction with the crystalline network. From the $^4I_{11/2}$ and $^4I_{13/2}$ levels, the Er^{3+} ions get excited to the $^4F_{7/2}$ and $^4F_{9/2}$ multiplets, respectively, by absorbing another 980 nm photon via excited state absorption (ESA). Further, non-radiative transitions from the $^4F_{7/2}$ level, populate the $^2H_{11/2}$, $^4S_{3/2}$ and $^4F_{9/2}$ multiplets, followed by radiative transitions, i.e., $^2H_{11/2} \rightarrow {}^4I_{15/2}$ (524 nm), $^4S_{3/2} \rightarrow {}^4I_{15/2}$ (554 nm) and $^4F_{9/2} \rightarrow {}^4I_{15/2}$ (655 nm). In addition, the $^4F_{7/2}$ level can also be populated by the energy transfer upconversion (ETU) process, in which the interaction of two or more nearby optically active ions is involved. The ETU channels in the case of Er^{3+}-doped samples are ETU1: ($^4I_{11/2}+{}^4I_{11/2} \rightarrow {}^4I_{15/2}+{}^4F_{7/2}$) and ETU2: ($^4I_{11/2}+{}^4I_{13/2} \rightarrow {}^4I_{15/2}+{}^4F_{9/2}$).

Both ESA and ETU processes can simultaneously coexist, although one of them usually dominates the upconversion process [39]. At higher concentrations of Er^{3+} ions, ETU generally dominates the ESA due to the shorter distance between the optically active ions [3]. In the case of co-doped samples, energy transfer (ET) from Yb^{3+} ions to Er^{3+} ions also occurs in addition to the GSA, ESA, ETU, non-radiative, and radiative transitions. By absorbing the 980 nm photons, Yb^{3+} ions are excited from $^2F_{7/2}$ to the $^2F_{5/2}$ level via GSA, and then they transfer their energy to the nearby ground-state Er^{3+} ions. This energy transfer can populate the excited $^2H_{11/2}$, $^4S_{3/2}$, and $^4F_{9/2}$ level followed by radiative transitions, as shown in Fig. 7. As the absorption cross-section of the Yb^{3+} ion is very high, the energy transfer causes the enhancement of the intensities of the green and red upconversion emissions in the co-doped samples.

Further, the observed increase in R/G ratio with respect to Er^{3+} or Yb^{3+} ion concentration in LuSGG nanocrystals could be due to the energy back transfer (EBT) from Er^{3+} to Yb^{3+} ion via ($^4S_{3/2}+{}^2F_{7/2} \rightarrow {}^4I_{13/2}+{}^2F_{5/2}$), ETU2 and cross-relaxation channels such as CR1: ($^2H_{11/2}$, $^4S_{3/2}+{}^4I_{15/2} \rightarrow {}^4I_{9/2}+{}^4I_{13/2}$) and CR2: ($^4F_{7/2}+{}^4I_{11/2} \rightarrow {}^4F_{9/2}+{}^4F_{9/2}$). In EBT processes, the mismatch of energy is around 1500 cm^{-1}, which can easily be bridged by two phonons of the host lattice (~760 cm^{-1}) [21]. These EBT, ETU2, and CR channels could inhibit the population of higher levels ($^2H_{11/2}$ and $^4S_{3/2}$), resulting in a suppression of the green emission [3,35,44]. At higher concentrations of Yb^{3+} ions, the EBT process is inherently efficient and becomes more prominent due to the reduced gap between Yb^{3+} and Er^{3+} ions and the increased Yb^{3+}/Er^{3+} ion ratio. Therefore, EBT, ETU2, and CR mechanisms enable an increase in the R/G ratio in the co-doped samples. Of course, in single-doped samples, ETU2 and CR channels play a key role in suppressing the green emission and increasing the R/G ratio. The energy transfer through CR channels is found to be stronger in the single-doped samples than in the co-doped samples (see Fig. 5(c)). Due to the addition of Yb^{3+} ions, the R/G ratio is reduced from 0.44 (LuSGG:1Er) to 0.26 (LuSGG:1Er1Yb). This could be due to the difference in enhancements of green and red emissions upon adding the Yb^{3+} ions to the LuSGG:Er host (see the inset of Fig. 5(b)), weak energy transfer through CR channels, and EBT processes. Because the EBT process further favours the ET processes by exciting the Yb^{3+} ions to higher levels, the $^2H_{11/2}$, $^4S_{3/2}$ and $^4F_{9/2}$ levels are more populated. In this way, the EBT processes form the loop. This could be the reason behind the relatively slow rise in the R/G ratio in the co-doped samples. However, the enhancement of the R/G ratio with respect to Yb^{3+} ion concentration in LuSGG:Er/Yb is observed to be high compared to that of LuGG:Er/Yb [3] and YGG:Er/Yb [19]. This indicates that there may be a higher transition probability of EBT or a higher CR2 rate or ETU2 rate in the LuSGG:Er/Yb sample relative to that

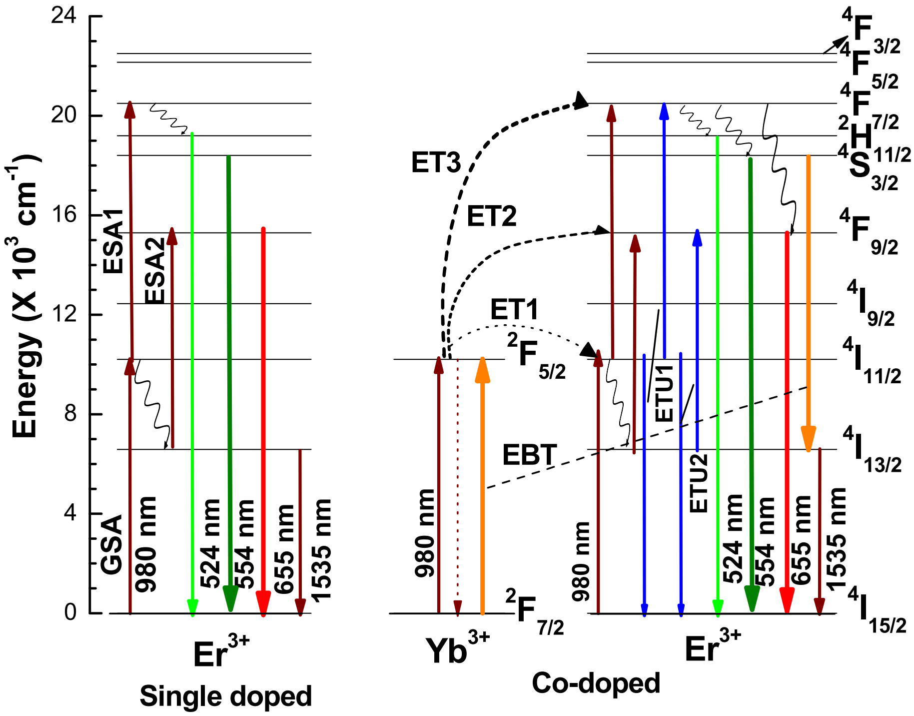

Fig. 7. Partial energy level diagram representing the upconversion mechanisms in both the single doped LuSGG:Er and co-doped LuSGG:Er/Yb samples.

in the LuGG:Er/Yb and YGG:Er/Yb samples. This would be better understood by considering the decay profiles of $^2H_{11/2}$, $^4S_{3/2}$ levels of Er^{3+} ions.

3.4. Decay curve analysis

The luminescence decay profiles for the $^2H_{11/2}$,$^4S_{3/2}$ level of Er^{3+} ions with respect to Er^{3+} or Yb^{3+} ion concentrations under 980 nm excitation were measured and are displayed in Fig. 8. All the decay curves show a non-exponential nature accompanied by the shortening of lifetimes. The average lifetime was calculated by using the following formula:

$$\tau_{eff} = \frac{\int t\, I\, dt}{\int I\, dt}$$

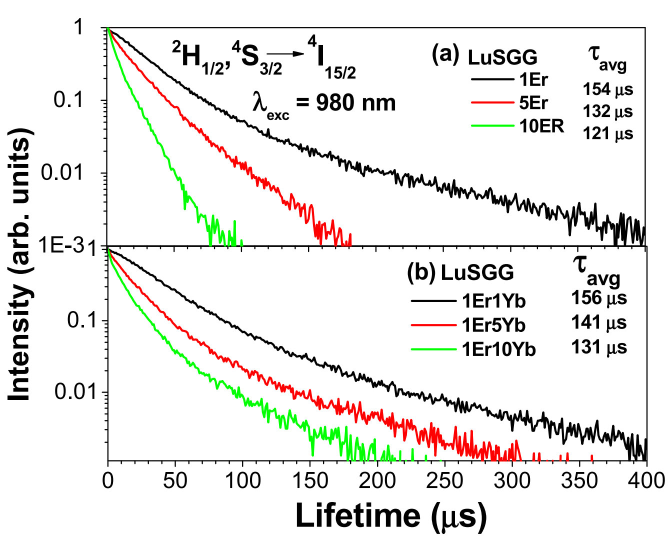

Fig. 8. The decay curves for the $^2H_{11/2}$, $^4S_{3/2}$ level of Er^{3+} ion in (a) LuSGG:Er and (b) LuSGG:Er/Yb nanocrystals.

The average lifetimes were found to be 155, 130, 120, 155, 140, and 130 μs for LuSGG:1Er, LuSGG:5Er, LuSGG:10Er, LuSGG:1Er1Yb, LuSGG:1Er5Yb, and LuSGG:1Er10Yb, respectively. It is noticed the lifetime of the $^2H_{11/2}$, $^4S_{3/2}$ levels is slightly enhanced with the addition of Yb^{3+} ions into the host, suggesting that energy was transferred from Yb^{3+} to Er^{3+} ions. However, with increasing Er^{3+} or Yb^{3+} ion concentrations, the lifetime values are quenched due to the increase in non-radiative energy transfer related to cross-relaxation processes at higher concentrations. The energy transfer sensitization efficiency (η_{ET}) can be evaluated using the following equation [45]:

$$\eta_{ET} = 1 - \frac{\tau_{Yb-Er}}{\tau_{Er}}$$

where $\tau_{Yb\text{-}Er}$ and τ_{Er} are the luminescence lifetimes of $^2H_{11/2}$, $^4S_{3/2}$ levels of Er^{3+} in Er/Yb co-doped LuSGG and Er-doped LuSGG samples, respectively. The calculated energy transfer efficiencies are 0, 0.097, and 0.161 for 1, 5, and 10 mol% of the Yb^{3+} ion in the LuSGG:Er/Yb sample, respectively. It is noticed that the energy transfer efficiencies increase gradually with Yb^{3+} ion concentration, indicating that the increased Yb^{3+} content improves the energy transfer probability from Yb^{3+} to Er^{3+} in the LuSGG host.

It is worth mentioning that the energy transfer due to CR channels is faster in the single-doped samples compared to the co-doped samples. This result is consistent with the slow rise of the R/G ratio in the co-doped samples. On the other hand, the shortening of lifetimes for the $^2H_{11/2}$, $^4S_{3/2}$ levels in co-doped systems may also be due to the increased EBT probability with increasing Yb^{3+} ion concentration, which greatly improves the non-radiative relaxation rate of the $^2H_{11/2}$, $^4S_{3/2}$, and thus decreases the average lifetimes [46]. Further, the lifetime values for the $^2H_{11/2}$, $^4S_{3/2}$ levels of the Er^{3+} ion in LuSGG:Er/Yb are observed to be higher compared to those of LuGG:Er/Yb [3] and YGG:Er/Yb [19], suggesting that the high value of the R/G ratio in LuSGG:Er/Yb is due to the higher ETU2 and CR2 rates but not due to the higher EBT probability.

3.5. CIE chromaticity

The basic requirements for the Plasma Display Panels (PDP) are emission colour purity and stability. The Commission International De I'Eclairage (CIE) chromaticity colour co-ordinates were evaluated with respect to the concentration of Yb^{3+} ions and pump power for the LuSGG:Er/Yb nanocrystals using the respective emission spectra. These co-ordinates are presented in Table 2 and pointed on the XY chromaticity co-ordinate graph shown in Fig. 9 for the study of the colour purity of the present nanocrystals. With increasing concentrations of Yb^{3+} ions, the co-ordinates shift from the standard green (0.300, 0.600) to the yellow colour, similar to the results observed in $SrGe_4O_9$:Er/Yb [47] and Lu_2O_3:Er/Yb [48] host matrices. However, a very slight variation in colour co-ordinates was observed with increasing pump power, and the colour co-ordinates fell into the green colour.

Table 2 also compares the colour co-ordinates for the other reported Er^{3+}/Yb^{3+} co-doped systems [45,49–52]. The colour purity (C_p) was calculated from the below equation [53]:

$$C_p = \frac{\sqrt{(x-x_i)^2+(y-y_i)^2}}{\sqrt{(x_d-x_i)^2+(y_d-y_i)^2}} x100\%$$

where x_i (0.33) and y_i (0.33) are the co-ordinates of the illuminated light, x_d and y_d are the dominant wavelength co-ordinates which were obtained by extending a straight line from the illuminated co-ordinates to the perimeter of the chromaticity diagram through the sample co-ordinates [53]. The obtained colour purity values for LuSGG:Er/Yb samples are closer to 100%, as given in Table 2. These values are high compared to those of $LaAlGe_2O_7$:Er phosphor [54] and Er-doped borate glass [4]. Hence, the present nanocrystals are suitable for the upconversion phosphors and white light emitting diodes as a green component under 980 nm laser excitation. Moreover, by adding the appropriate amount of Tm^{3+} ions to the LuSGG:1Er10Yb host, it is possible to harness the white light.

3.6. NIR-to-NIR luminescence

Fig. 10 represents the NIR emission spectra of LuSGG:Er/Yb nanocrystals under 980 nm excitation with varying Er^{3+} or Yb^{3+} ion concentrations. The spectra consist of a single band centred at 1535 nm associated with the $^4I_{13/2} \rightarrow {}^4I_{15/2}$ transition of the Er^{3+} ion. As can be seen from Fig. 7, different routes for the population of the $^4I_{13/2}$ level take place. In the case of single-doped LuSGG:Er nanocrystals, the Er^{3+} ions from the ground state $^4I_{15/2}$ level get excited to $^4I_{11/2}$ level by absorbing a 980 nm photon (GSA). From there, part of the energy is relaxed non-radiatively to the $^4I_{13/2}$ level, followed by the radiative transition $^4I_{13/2} \rightarrow {}^4I_{15/2}$ that leads to the characteristic emission at 1535 nm. Likewise, in the co-doped LuSGG:Er,Yb nanocrystals, in addition to the GSA, resonant energy transfer from Yb^{3+} to Er^{3+} takes

Table 2
CIE colour co-ordinates and colour purity values in LuSGG:Er/Yb nanocrystals and other reported Er^{3+}/Yb^{3+}-doped systems.

LuSGG:Er/Yb (in mol %)	Colour co-ordinates		Colour purity	Ref.
	x	y		
1Er1Yb	0.337	0.654	95	Present work
1Er5Yb	0.367	0.622	97	
1Er10Yb	0.437	0.552	97	
At high pressure (113–1020 mW)				
1Er1Yb	0.33	0.66	–	
Reported values				
$SrGe_4O_9$:Er/Yb	0.347	0.606	–	[46]
Gd_2O_3:Er/Yb	0.23	0.52	–	[49]
Sr_2CeO_4:Er/Yb	0.32	0.66	–	[50]
YBO_3:Er/Yb	0.29	0.64	–	[51]
$BaLaAlO_4$:Er/Yb	0.29	0.68		[52]

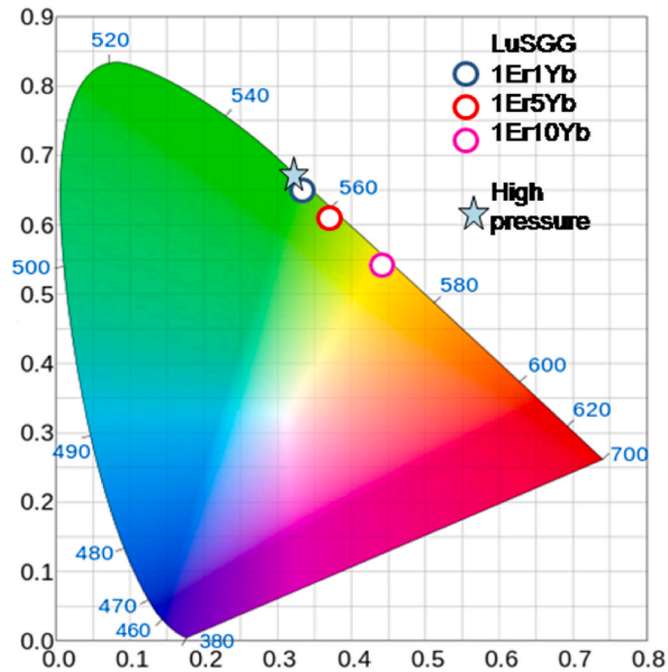

Fig. 9. CIE chromaticity diagram of LuSGG:Er/Yb nanocrystals.

place (ET1), which populates the $^4I_{11/2}$ level and then the $^4I_{13/2}$ level. In both cases, the non-radiative energy transfer from the $^4S_{3/2}$ level may also populate the $^4I_{13/2}$ level. As can be seen from the inset of Fig. 10(b), a 3-fold enhancement has been observed in the NIR emission intensity with the addition of Yb^{3+} ions into the LuSGG:Er nanocrystals. This confirms the energy transfer from Yb^{3+} to Er^{3+} ions in the LuSGG nanocrystals. From Fig. 10, it is also noticed that the emission intensity increases up to 5 mol% of Er^{3+} or Yb^{3+} ion concentrations and then decreases. This could be due to the increased probability of ETU2 at higher Er^{3+} or Yb^{3+} concentrations (i.e., beyond 5 mol%). This result is also consistent with the increase of red emission intensity and the increase in R/G ratio with increasing concentration. In addition, at higher concentrations, the energy of Ln^{3+} ions may migrate to defects or other quenching centres, finally resulting in a fast decrease in luminescence intensity. A different emission profile was observed for the 10 mol% of Er^{3+}/Yb^{3+} ions in the LuSGG host, which could be due to radiation trapping effects. In this process, a part of the photons emitted by an Er^{3+} ion are reabsorbed by a non-excited Er^{3+}, inducing the $^4I_{13/2} \rightarrow {}^4I_{15/2}$ transition and the subsequent emission. Actually, the reabsorption is more efficient at around 1530 nm than at other wavelengths due to the spectral overlap of absorption and emission, and as a result, the peak intensity decreases [55].

4. Conclusions

In summary, LuSGG:Er and LuSGG:Er/Yb nanocrystals were synthesised by the citrate sol-gel method. The nanocrystals exhibited a cubic garnet phase with an average crystallite size of 27 nm and the morphology of irregular grains with agglomeration. Diffuse reflectance and excitation spectra have shown characteristic absorption bands, which were used to generate the energy level diagram. NIR to visible upconverted luminescence spectra were obtained by exciting the samples with a 980 nm laser diode. The spectra present two strong bands in the green and red regions, which correspond to $^2H_{11/2}$, $^4S_{3/2} \rightarrow {}^4I_{15/2}$ and $^4F_{9/2} \rightarrow {}^4I_{15/2}$, respectively. With the incorporation of Yb^{3+} ions into the LuSGG:Er sample, the luminescence intensity was enhanced due to the presence of energy transfer from Yb^{3+} to Er^{3+} ions. The effect of concentration and pump power on the upconverted luminescence properties was studied. The R/G ratio increased significantly with

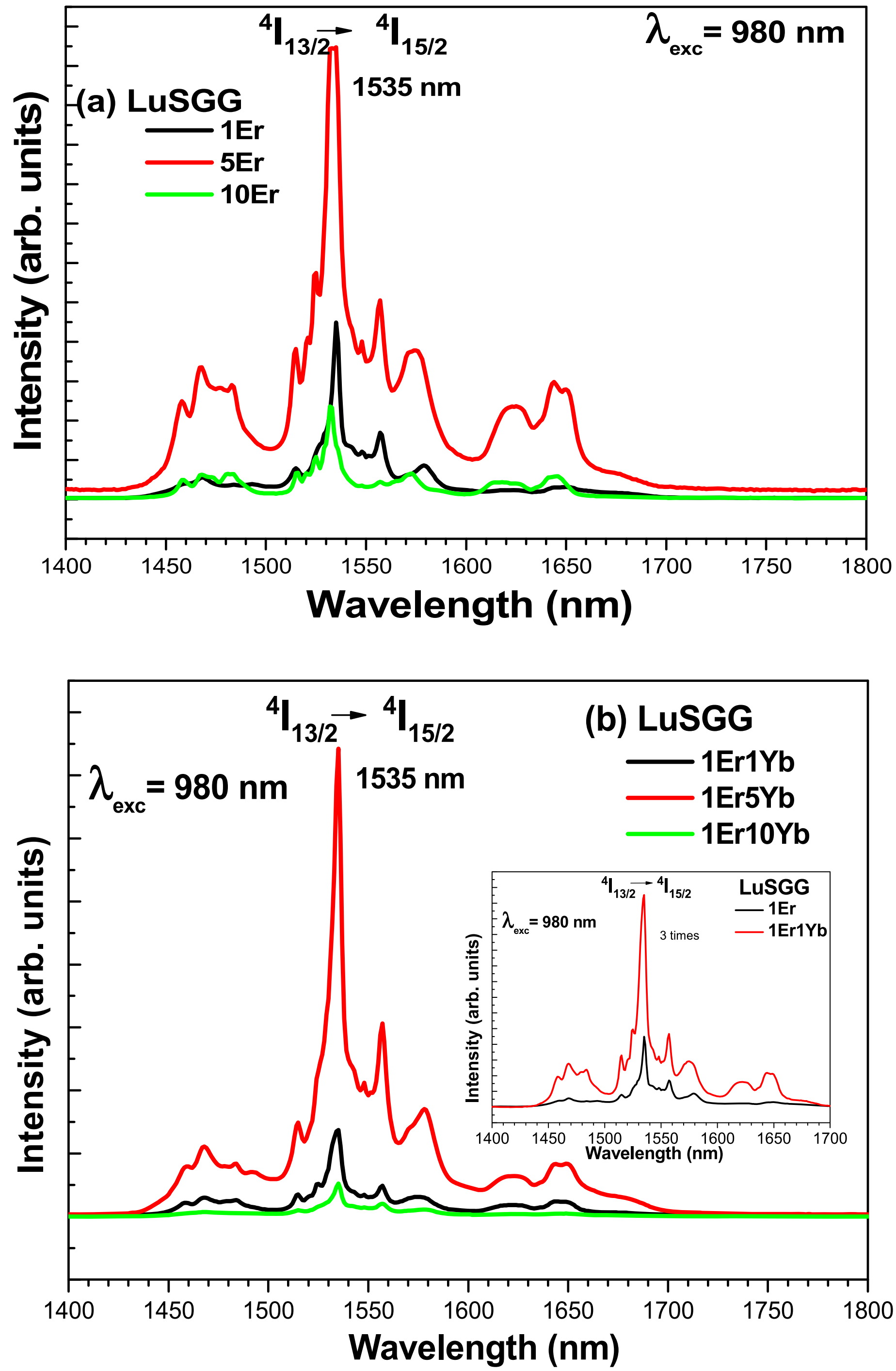

Fig. 10. NIR emission spectra of (a) LuSGG:Er and (b) LuSGG:Er/Yb nanocrystals under 980 nm excitation.

increasing Er^{3+} or Yb^{3+} ion concentrations, while it wasn't varied with respect to pump power. The enhancement of the R/G ratio with concentration was due to the energy back transfer from $Er^{3+}\rightarrow Yb^{3+}$ and cross-relaxation channels present in the sample. The higher R/G values in the present host suggested that it is easy to tune the emission colour with concentrations of Er^{3+} or Yb^{3+} ions. There were two photons involved in the upconversion mechanism. The decay curves for the $^2H_{11/2}$, $^4S_{3/2}$ levels of the Er^{3+} ion displayed a non-exponential nature. The observed decrease in the lifetime values with increasing concentration was due to concentration quenching via cross-relaxations and energy back transfer. CIE chromaticity co-ordinates were located in the green region and were shifted towards the yellow region with increasing Yb^{3+} concentration, but they were not changed with the pump power. NIR to NIR emission spectra displayed the intense band at 1535 nm ascribed to the $^4I_{13/2} \rightarrow {}^4I_{15/2}$ transition of the Er^{3+} ion, whose intensity was raised with the addition of the Yb^{3+} ion due to energy transfer from Yb^{3+} to Er^{3+}. The intensity also varied with the concentration, and optimum concentration was observed for 5 mol%. The results showed that LuSGG: Er and LuSGG:Er/Yb phosphors could be good candidates for light emitting diodes, PDP, and display applications.

CRediT authors statement

N. Rama Devi: Formal analysis,Writing-Original draft, **R. Praveena**:

Preparation of samples, Resources, Conceptualization, Methodology, Validation, Writing-review & editing, Supervision, **V. Venkatramu**: Resources, measurement, Data curation, Writing-review & editing, **Ch. B. Basavapoornima**: Investigation, Data curation, Measurement, **V. Lavin**: Resources, Measurement, Validation, Writing-review & editing, **B.D. Joshi**: Writing-review & editing, Validation, **N.S.S.V. Raja Rao**: Preparation of samples, Investigation, Formal analysis.

Author statement

All the authors agree the submission, and would like to state that: the article is original, has been written by the stated authors who are all aware of its content and approve its submission, has not been published previously, it is not under consideration for publication elsewhere, no conflict of interest exists, the article will not be published elsewhere in the same form, in any language, without the written consent of the publisher.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

Dr. R. Praveena is grateful to the Gayatri Vidya Parishad College of Engineering (A), Visakhapatnam, for the partial financial support. Dr. V. Venkatramu is grateful to DST, New Delhi for the sanction of research project (No. INT/PORTUGAL/P-04/2017) under India-Portugal Bilateral Scientific and Technological Cooperation. Dr. V. Lavín thanks Ministerio de Ciencia e Innovación (MICIIN) for the Project under the National Program of Sciences and Technological Materials (PID2019-106383 GB-C44) and EU-FEDER funds. The authors are also thankful to Prof. C.K. Jayasankar, Sri Venkateswara University (Tirupati, India) for providing few characterization facilities.

References

[1] A. Dwivedi, K. Mishra, S.B. Rai, J. Appl. Phys. 120 (2016), 043102.
[2] J. Yang, C. Zhang, C. Peng, C. Li, L. Wang, R. Chai, J. Lin, Chem. Eur J. 15 (2009) 4649.
[3] M. Rathaiah, P. Haritha, K. Linganna, V. Monteseguro, I.R. Martin, A.D. Lozano-Gorrin, P. Babu, C.K. Jayasankar, V. Lavin, V. Venkatramu, ChemPhysChem 16 (2015) 3928.
[4] F. Rivera-Lopez, M.E. Torres, G. Gil de Cos, Mat. Today Commun. 27 (2021), 102434.
[5] Z. Pan, A. Ueda, R. Mu, S.H. Morgan, J. Lumin. 126 (2007) 251.
[6] B. Schaudel, P. Goldner, M. Prassas, F. Auzel, J. Alloys Compd. 300 (2000) 443.
[7] S. Sinha, K. Kumar, Opt. Mater. 75 (2018) 770.
[8] A. Kumar, S.P. Tiwari, H.C. Swart, J.C.G. Esteves da Silva, Opt. Mater. 92 (2019) 347.
[9] S.K. Maurya, J.C.G. Esteves da Silva, M. Mohan, R. Poddar, K. Kumar, J. Alloys and Compds. 865 (2021), 158737.
[10] P.P. Mokoena, D.O. Oluwole, T. Nyokong, H.C. Swart, O.M. Ntwaeaborwa, Sens. Actu. A 331 (2021), 113014.
[11] K. Pavani, J.P.C. do Nascimento, S.K. Jakka, F.F. do Carmo, A.J.M. Sales, M. J. Soares, M.P.F. Graca, F.J.A. de Aquino, D.X. Gouveia, A.S.B. Sombra, J. Lumin. 235 (2021), 117992.
[12] M. Rathaiah, I.R. Martin, P. Babu, K. Linganna, C.K. Jayasankar, V. Lavin, V. Venkatramu, Opt. Mater. 39 (2015) 16.
[13] R. Krsmanovic, V.A. Morozov, O.I. Lebedev, S. Polizzi, A. Spegini, M. Bettinelli, G. Van Tendeloo, Nanotech 18 (2007), 325604.
[14] H.J. Byun, W.S. Song, Y.S. Kim, H. Yang, J. Phys. D 43 (2010), 195401.
[15] S. Li, Y. He, M. Li, F. Zeng, X. Wang, C. Sun, Z. Su, J. Alloys and Compds. 794 (2019) 227.
[16] G. Chen, H. Qiu, P.N. Prasad, X. Chen, Chem. Rev. 114 (2014) 5161.
[17] S.F. Leon-Luis, V. Monteseguro, U.R. Rodriguez-Mendoza, M. Rathaiah, V. Venkatramu, A.D. Lozano-Gorrin, R. Valiente, A. Munoz, V. Lavin, RSC Adv. 4 (2014), 57691.
[18] A.D. Lozano-Gorrin, U.R. Rodriguez-Mendoza, V. Venkatramu, V. Monteseguro, M. A. Hernandez-Rodriguez, I.R. Martin, V. Lavin, Opt. Mater. 84 (2018) 46.
[19] V. Monteseguro, V. Venkatramu, U.R. Rodriguez-Mendoza, V. Lavin, Dalton Trans. 50 (2021) 9512.
[20] F. Vetrone, J.C. Boyer, J.A. Capobianco, A. Speghini, M. Bettinelli, J. Phys. Chem. B 106 (2002) 5622.
[21] V. Venkatramu, M. Giarola, G. Mariotto, S. Enzo, S. Polizzi, C.K. Jayasankar, F. Piccinelli, M. Bettinelli, A. Speghini, Nanotech 21 (2010), 175703.
[22] A.A. Kaminskii, G. Boulon, M. Buoncristiani, B. Dibartolo, A. Kornienko, V. Mironov, Phys. Status Solidi, A Appl. Res. 141 (1994) 471.
[23] K. Wu, L.Z. Hao, H.J. Zhang, H.H. Yu, H.J. Cong, J.Y. Wang, Opt Commun. 284 (2011) 5192.
[24] Y.N. Xu, W.Y. Ching, B.K. Brickeen, Phys. Rev. B61 (2000) 1817.
[25] D. Li, W. Qin, S. Liu, W. Pei, Z. Wang, P. Zhang, J. Alloys and Compds. 653 (2015) 304.
[26] S. Sakirzanovas, A. Kareiva, Lith. J. Phys. 47 (2007) 75.
[27] R. Praveena, P. Venkatalakshmamma, V. Sravani Sameera, V. Venkatramu, I. R. Martín, V. Lavín, C.K. Jayasankar, Mat. Res. Bull. 101 (2018) 347.
[28] V. Monteseguro, M. Rathaiah, K. Linganna, A.D. Lozano-Gorrin, M.A. Hernandez-Rodriguez, I.R. Martin, P. Babu, U.R. Rodriguez-Mendoza, F.J. Manjon, A. Munoz, C.K. Jayasankar, V. Venkatramu, V. Lavin, Opt. Mater. Exp. 5 (2015) 1661.
[29] B. Malysaa, A. Meijerinkb, T. Justela, J. Lumin. 202 (2018) 523.
[30] B.M. Walsh, G.W. Grew, N.P. Barnes, J. Phys. Chem. Solids 67 (2006) 1567.
[31] V. Monteseguro, M. Rathaiah, K. Linganna, A.D. Lozano-Gorrin, M.A. Hernandez-Rodriguez, I.R. Martin, P. Babu, U.R. Rodriguez-Mendoza, F.J. Manjon, A. Munoz, C.K. Jayasankar, V. Venkatramu, V. Lavin, Opt. Mat. Exp. 5 (2015) 1661.
[32] K.W. Powers, M. Palazuelos, B.M. Moudgil, S.M. Roberts, Nanotoxicology 1 (2007) 42.
[33] L.F. Hakim, J.L. Portman, M.D. Casper, A.W. Weimer, Powder Technol. 160 (2005) 149.
[34] V. Singh, M. Seshadri, N. Singh, M. Mohapatra, J. Mat. Sci.: Mat. in Electr. 30 (2019) 2927.
[35] V. Singh, P. Haritha, V. Venkatramu, S.H. Kim, Spectrochim. Acta, Part A 126 (2014) 306.
[36] R. Praveena, P. Venkatalakshmamma, V. SravaniSameera, V. Venkatramu, I. R. Martín, V. Lavín, C.K. Jayasankar, Mat. Res. Bull. 101 (2018) 347.
[37] N. Ramadevi, R. Praveena, V. Venkatramu, Ch Basavapoornima, V. Lavin, B. D. Joshi, J. Kor. Phys. Soc. 81 (2022) 991.
[38] B.G. Wybourne, Spectroscopic Properties of Rare Earths, Wiley-Interscience, New York, 1965.
[39] V. Venkatramu, S.F. Leon-Luis, U.R. Rodriguez-Mendoza, V. Monteseguro, F. J. Manjon, A.D. Lozano-Gorrin, R. Valiente, D. Navarro-Urrios, C.K. Jayasankar, A. Munoz, V. Lavin, J. Mater. Chem. 22 (2012), 13788.
[40] S. Liu, H. Ming, J. Cui, S. Liu, W. You, X. Ye, Y. Yang, H. Nie, R. Wang, J. Phys. Chem. C 122 (2018), 16289.
[41] Y. Lei, H. Song, L. Yang, L. Yu, Z. Liu, G. Pan, X. Bai, L. Fan, J. Chem. Phys. 17 (2005) 123.
[42] M. Pollnau, D.R. Gamelin, S.R. Lüthi, H.U. Güdel, Phys. Rev. B 61 (2000) 3337.
[43] V. Mahalingam, N. Rafik, F. Vetrone, J.A. Capobianco, Opt. Exp. 20 (2012) 111–119.
[44] G. Chen, G. Somesfalean, Y. Liu, Z. Zhang, Q. Sun, F. Wang, Phys. Rev. B 75 (2007), 195204.
[45] J.A. Mares, N. Wenjiang, G. Boulon, J. Phys. France 51 (1990) 1655.
[46] H. Suo, C. Guo, T. Li, J. Phys. Chem. C 120 (2016) 2914.
[47] D. Chavez, C.R. Garcia, J. Oliva, E. Montes, A.I. Mtz-Enriquez, M.A. Garcia-Lobato, L.A. Diaz-Torres, Cer. Int. 45 (2019), 16911.
[48] P. Gluchowski, L. Marciniak, M. Lastusaari, W. Strek, J. Alloys and Compds. 799 (2019) 481.
[49] P.K. Yadaw, R.K. Padhi, V. Dubey, M.C. Rao, N. Kumar Swamy, Inorg. Chem. Commun. 143 (2022), 109736.
[50] E. Viesca-Villanueva, J. Oliva, D. Chavez, C.M. Lopez-Badillo, C. Gomez-Solis, A. I. Mtz-Enriquez, C.R. Garcia, J. Phys. Chem. Solids 145 (2020), 109547.
[51] J. Oliva, D. Chavez, A. Gonzalez-Galvan, E. Viesca-Villanueva, L.A. Diaz-Torres, J. Fraga, C.R. Garcia, Optik 241 (2021), 167011.
[52] E. Fred Schubert, Light Emitting Diodes, second ed., Cambridge University Press, 2006, pp. 300–301 (Chapter 17).
[53] Y.C. Li, Y.H. Chang, Y.F. Lin, Y.J. Lin, Y.S. Chang, Appl. Phys. Lett. 89 (2006) 5.
[54] N.S. Prabhu, A.N. Meza-Rocha, O. Soriano-Romero, U. Caldino, E.F. Huerta, C. Falcony, M.I. Sayyed, H. Al-Ghamdi, A.H. Almuqrin, S.D. Kamath, J. Lumin. 238 (2021), 118216.
[55] X. Feng, S. Tanabe, T. Hanada, J. Am. Ceram. Soc. 84 (2001) 165.